Michael Liu

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Education

University of Waterloo — *Sept 2025 - Apr 2030*
BMath in Mathematics

- Major Average: 94% | Cumulative Average: 93%, Student ID: 21177966
- Activities: Varsity Cross-Country, Varsity Track, WAT.ai (Artificial Intelligence) Design Team

Job Experience

Machine Learning Research Assistant — *Toronto, ON*
The Hospital for Sick Children (SickKids) — *May 2026 - Aug 2026*

- Incoming Summer 2026 | Machine Learning for Drug Discovery

Machine Learning Engineer — *Waterloo, ON*
WAT.ai — *Sept 2025 - Feb 2026*

- Collaborated with a small team of students on an applied medical imaging project focused on improving the segmentation of microaneurysms in fundus imaging.
- Led dataset curation and preprocessing, implementing reproducible data pipelines for data splits and augmentation to support ablation studies.
- Co-authored a research paper detailing our methodology and results.

Tennis Instructor — *Unionville, ON*
Unionville Tennis Club — *Apr 2022 - Aug 2025*

- TPA Certified Instructor | 500+ Hours On-Court

Projects

ArXtract: AI Search Engine for ML Research @ CxC Hackathon — *arxtract-cxc.vercel.app*

- Built a full-stack research tool (React + TypeScript frontend, FastAPI backend) for analyzing machine learning papers through user prompts and arXiv inputs.
- Implemented embedding-based retrieval using cosine similarity between user research queries and paper abstracts/paragraphs to rank relevant sections of arXiv papers.
- Built a retrieval-augmented chatbot that answers research questions by ranking paper chunks using embedding similarity and responding using the top-scoring sections.
- Engineered structured extraction of key ML fields (task type, problem, contribution, datasets) to standardize paper summaries.

Enchancing Microaneurysm Segmentation in Retinal Fundus Imaging @ WAT.ai — *github/fundus-image-segmentation*

- Built a PyTorch data pipeline with image-mask augmentations for fundus imaging datasets.
- Designed and trained (via SSH) HydraLA-Net (U-Net Variation) models for the semantic segmentation of microaneurysms, hemorrhage, soft and hard exudates from scratch in PyTorch.
- Conducting ablation studies on applications of contrast enchancement preprocessing and loss function selection to improve small-lesion segmentation performance.

Technical Skills

Programming Languages:	Python, SQL, C, TypeScript
Data/ML:	PyTorch, scikit-learn, Pandas, NumPy, Matplotlib, LLM/GenAI
Databases/Tools:	PostgreSQL, MySQL; Git, GitHub; Jupyter Notebook, VS Code; Excel, Tableau
Frontend:	React, HTML, CSS